

SΔφ-28 — Default Power as Low-Cost Path

Assignment: Minimal Conditions for Invisible Fixation after SΔφ-27 (v1.0)

SΔφ Series / Sofience-Δφ Formalism

Abstract. This paper extends SΔφ-27 by proposing default power as a minimal political and infrastructural mechanism of path fixation. Where older forms of power operated through prohibition, command, and punishment, default power operates by assigning one path as the cheapest continuation while leaving formal alternatives intact. The central claim is that power increasingly functions not by closing paths through force, but by redistributing operational cost so that one continuation becomes frictionless and others become obscure, costly, or practically ineditable. In this sense, SΔφ-28 translates the dynamics of low-cost convergence into a theory of power: what appears as convenience at the level of action may function as fixation at the level of structure.

1. From command to default

Classical power appears as prohibition. It tells a system what must not be done, attaches punishment to forbidden paths, and attempts to block deviation by visible force. Default power works differently. It does not need to abolish alternatives. Instead, it arranges one continuation as the cheapest operational path, leaving other paths formally available while making them frictional, obscure, or practically expensive. A default is therefore not a neutral setting. It is a low-cost assignment of continuation.

Default does not abolish alternatives; it governs by making one path the cheapest continuation.

2. Why default is stronger than prohibition

A prohibition invites resistance, evasion, and explicit contest. A default often avoids these reactions because it preserves the experience of voluntary choice. The subject still feels that selection remains open, but repeated use of the default path lowers operational cost, increases repetition, and gradually raises the restoration cost of alternatives. What begins as convenience becomes fixation. The power of the default lies precisely in the fact that it can govern without appearing to command.

3. Minimal axioms

Axiom 1. Power increasingly operates not by prohibition, but by default path assignment.

Axiom 2. A default path is the continuation with the lowest operational cost among available paths.

Axiom 3. Repeated default selection intensifies path fixation.

Axiom 4. As fixation increases, restoration cost across non-default alternatives rises.

Axiom 5. Formal choice may remain intact while practical editability shrinks.

4. Minimal formalization

Let P be a set of available paths. Let $C_{op}(P_i)$ denote the operational cost of path P_i . The default path is defined as $P_{default} = \operatorname{argmin} C_{op}(P_i)$. If repetition of that path is represented by $R(P_{default})$, and fixation by $F(P_{default})$, then repeated low-cost

continuation implies an increase in fixation: $R(P_{\text{default}})$ up, therefore $F(P_{\text{default}})$ up. As fixation rises, restoration cost across alternatives also rises: $F(P_{\text{default}})$ up, therefore $C_{\text{rest}}(P_{\text{alt}})$ up. The political significance of the default lies in this sequence: the cheapest continuation becomes the least visible form of command.

5. Invisible fixation

Invisible fixation occurs when a path continues to appear optional while in practice becoming increasingly difficult to revise. This is why default power is more stable than coercive power. The subject experiences continuity rather than enforcement, optimization rather than domination, convenience rather than closure. Yet the structural result may be stronger than prohibition: alternatives remain nominally available but cease to function as realistic routes of action.

The new command is not “you must,” but “this is easiest.”

6. Platforms, interfaces, and AI

Infrastructural systems such as interfaces, recommendation engines, protocol standards, and platform defaults are now major sites of power because they assign path cost before explicit decision takes place. The decisive shift introduced by AI is not simply that it can choose instead of humans, but that it can increasingly edit the cost landscape within which humans choose. This moves power away from visible institutions of command and toward systems that continuously shape convenience, attention, friction, and path continuation.

7. Freedom under default power

Under default power, freedom cannot be measured merely by counting formal options. A system may present many alternatives while preserving only one cheap continuation. Freedom therefore depends less on nominal choice than on whether non-default paths remain practically editable. This marks the transition from a moral language of choice to a structural language of editability.

8. Conclusion

SΔφ-28 proposes that default is the new command. Power no longer needs to close paths by explicit force if it can make one path the cheapest continuation. Repeated default selection converts convenience into fixation, and fixation into restoration cost explosion. The next problem is therefore no longer who commands, but who can still edit the default. In this sense, SΔφ-28 prepares the transition to a theory of editable defaults and post-prohibitive governance.

Series references.

SΔφ-26 — Irreversibility as Restoration Cost Explosion: Minimal Fixation Conditions after Pre-detection (v1.0). DOI: 10.5281/zenodo.18818936

SΔφ-27 — Low-Cost Convergence and Restoration Cost Explosion: Exception Compression as a Minimal Fixation Mechanism (v1.0). DOI: 10.5281/zenodo.18822918